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FINAL REPORT

BUSINESS INTELLIGENCE

TOPIC: BUSINESS REVENUE ANALYSIS

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CHAPTER 1: PROJECT INTRODUCTION

1.1. Background and necessity of the project

In the context of a rapidly developing digital economy, data is considered the "oil of the 21st century." Businesses are generating enormous amounts of data from various sources, but much of it remains untapped and unanalyzed. Business Intelligence (BI) is a set of methods, processes, architectures, and technologies that enable the transformation of raw data into meaningful information, helping businesses make strategic decisions based on factual evidence.

Masan Group is one of the largest private economic groups in Vietnam, primarily operating in the fast-moving consumer goods (FMCG) sector. With a

nationwide distribution network across traditional, modern, and digital channels, Masan processes tens of thousands of transactions daily. Marketing data is collected from numerous touchpoints in the customer journey, from brand awareness (Impression) to purchase conversions (Purchase), creating a rich but fragmented and difficult-to-exploit dataset.

The challenge was how to effectively integrate, clean, and analyze data from six distribution channels and five stages in the marketing conversion funnel, thereby providing a comprehensive view of marketing effectiveness for management. This is the reason behind the creation of this BI system development project.

1.2. Project Objectives

The project sets out the following specific objectives:

- Develop an automated ETL (Extract-Transform-Load) process to reliably and repeatably extract, clean, and load data from a source Excel file into a PostgreSQL data warehouse system.
- The data warehouse is designed using the Star Schema model, with dimension and fact tables suitable for multidimensional marketing analytics requirements.
- Develop interactive dashboards in Power BI that display key KPIs and allow for in-depth analysis from multiple perspectives.
- Containerize the entire database system using Docker to ensure consistency and easy deployment across multiple environments.
- Provides a comprehensive overview of marketing effectiveness by channel, campaign, conversion funnel, and progress toward goals.

1.3. Scope of the study

1.3.1. Data Scope

The project utilizes Masan Group's marketing dataset, including transactions from the 2023-2024 period. The data is stored in an Excel file (masan_case.xlsx) with 37 attributes, covering all aspects from revenue, costs, and profits to specialized marketing metrics such as Marketing ROI, Market Share, and Conversion Proxy.

1.3.2. Technical Scope

Technically, the project is deployed locally using Docker containers for PostgreSQL, Python for ETL processing, and Power BI Desktop for visualization. The system is designed with a three-layer architecture: Source Layer, Storage Layer, and Presentation Layer.

1.3.3. Limits

The project does not include building real-time APIs, integrating with existing ERP or CRM systems, or implementing Machine Learning predictive models. This is a descriptive BI platform focused on analyzing historical data.

1.4. Report Structure

The report is organized into eight chapters. Chapter 1 provides an overview of the project. Chapter 2 describes the source dataset in detail. Chapter 3 presents the overall architecture of the BI solution. Chapter 4 delves into the data warehouse design using the Star Schema model. Chapter 5 describes the ETL process in detail. Chapter 6 presents the design and components of the Power BI dashboard. Chapter 7 analyzes and discusses the results. Finally, Chapter 8 concludes and suggests future development directions.

CHAPTER 2: DATA OVERVIEW

2.1. Data Sources

The data used in this project is taken from the Microsoft Excel file `masan_case.xlsx`, which is a comprehensive dataset on Masan Group's marketing and sales activities. This file contains a single sheet (Sheet1) with a flat table structure, where each row represents a transaction/marketing activity, and each column is an attribute describing that transaction.

The dataset includes information from multiple distribution channels (Facebook, Ecommerce, TV, Retail promotion, Traditional trade, Other), 5 stages in the marketing funnel (Impression, Website visit, Click, Add to cart, Purchase),

multiple geographic regions nationwide, and dozens of detailed financial and operational metrics.

2.2. Dataset Description

Table 2.1 details the 37 attributes (columns) in the source dataset:

Table 2.1 – Description of the attributes of the masan_case.xlsx dataset

Attribute name	Describe	Data type	Note
OrderID	Order identifier	BIGINT	The primary, unique key
Date	Date the transaction occurred	DATE	YYYY-MM-DD format
Year / Month	Year / Month	INT	Derived from Date
Region	Geographic region	TEXT	Southern, Central, and Northern regions
Branch	Branch	TEXT	Ho Chi Minh City, Hanoi, Binh Duong...
Department	Department in charge	TEXT	Operations, Marketing, Sales...
Product	Product name	TEXT	Fresh milk has very little sugar, etc.
Category	Product catalog	TEXT	Fresh milk, yogurt...
Stage	funnel stage	TEXT	Impression → Purchase
Channel	Distribution channel	TEXT	Facebook, Ecommerce, TV...
Information	Source/Type of Information	TEXT	Facebook orders, complaints...
Machine	Production machine/line	TEXT	A, B, C
Quantity	Quantity of products	NUMERIC	Unit of product
Revenue	Revenue	NUMERIC	Thousand Vietnamese Dong (VND)

Cost	Total cost	NUMERIC	Thousand Vietnamese Dong (VND)
Profit	Gross profit	NUMERIC	Revenue – Cost
ProfitMargin	Profit margin	NUMERIC	Profit / Revenue (0–1)
InventoryLevel	Inventory	NUMERIC	Number of units in stock
Calls	Number of customer service calls	NUMERIC	Call/period
WaitingTime	Waiting time	NUMERIC	Minute
Budget	Marketing budget	NUMERIC	Thousand Vietnamese Dong (VND)
Target	Revenue targets	NUMERIC	Thousand Vietnamese Dong (VND)
RawMaterialCost	Raw material costs	NUMERIC	Thousand Vietnamese Dong (VND)
LaborCost	Labor costs	NUMERIC	Thousand Vietnamese Dong (VND)
LogisticsCost	Shipping costs	NUMERIC	Thousand Vietnamese Dong (VND)
MarketingCost	Marketing costs	NUMERIC	Thousand Vietnamese Dong (VND)
UnitCost	Unit cost	NUMERIC	VND/product
RevenuePerUnit	Unit revenue	NUMERIC	VND/product
ROI_Marketing	ROI marketing	NUMERIC	Marketing return on investment
ConversionProxy	Conversion rate	NUMERIC	Proxy cho conversion rate

CostPerMachineUnit	Cost/machine	NUMERIC	Cost per unit of machine
RegionProfitShare	Profit share by region	NUMERIC	Ratio 0–1
MarketSize	Market size	NUMERIC	Thousand Vietnamese Dong (VND)
MarketShare	Market share	NUMERIC	Ratio 0–1
CustomerSegment	Customer segmentation	TEXT	Modern channels, e-commerce...
PromotionCampaign	Promotional campaign	TEXT	Year-End, Tet, Summer...

Revenue

2.3.1. Distribution Channels

The data covers six main distribution channels: Facebook (social media), Ecommerce, TV, Retail promotion, Traditional trade, and Other. Each channel has its own characteristics in terms of cost, conversion rate, and effectiveness at each stage of the funnel.

2.3.2. Revenue Funnel

The data reflects the five stages of the digital marketing funnel in sequence: Impression (Brand Awareness) → Website Visit → Click → Add to Cart → Purchase. Tracking data at each stage allows for analysis of dropout rates and optimization of each step in the customer journey.

2.3.3. Revenue Campaign

Marketing campaigns are categorized into four groups: Seasonal (Tet, Spring, Summer, Mid-Autumn), Year-End, Back-to-School, and Other. This classification helps businesses evaluate the effectiveness of each campaign type and plan budgets for subsequent periods.

2.3.4. Data Quality

The source data presented several quality challenges that needed to be addressed during the ETL phase: (1) Inconsistent column names across data versions (e.g., "roi marketing", "ROI_Marketing", "roimarketing" all refer to the same metric); (2) Mixed date formats (ISO yyyy-mm-dd and dd/mm/yyyy); (3) Missing values in some optional fields; (4) Duplicate data by OrderID. All of these issues were thoroughly addressed in the Transform step of the ETL pipeline.

CHAPTER 3: BUSINESS INTELLIGENCE SOLUTION ARCHITECTURE

3.1. Architectural Overview

The BI system architecture in this project follows the industry-standard 3-layer model, including: Data Source Layer, Storage & Processing Layer, and Presentation Layer. This model is widely used in enterprise BI systems due to its clarity, ease of maintenance, and high scalability.

The data flow is designed to be unidirectional, ensuring consistency and control: data only flows from the source through the staging area into the data warehouse, never in the reverse direction. Each layer has a clear role and defined responsibilities.

3.1.1. Source Layer

This layer contains raw, unprocessed data. Within the project scope, the sole data source is the Microsoft Excel file `masan_case.xlsx` with 37 columns and thousands of marketing transaction records. This is "as-is" data from the operating system, which may contain inconsistencies, missing values, and non-standard formatting.

3.1.2. Staging Layer

The staging area is an intermediate buffer zone in a PostgreSQL database, the staging schema. Data is loaded here with a TEXT structure for all columns, retaining the original column names from Excel. The purpose of this layer is to separate the

extraction of source data from the transformation process, allowing ETL reruns without affecting the data warehouse. Each time the ETL is run, the staging table is TRUNCATED and reloaded entirely.

3.1.3. Data Warehouse

The Data Warehouse is located within the PostgreSQL `dw` schema and is designed according to the Star Schema model with eight dimension tables and one central fact table. This layer provides cleaned, normalized, and optimally organized data for analysis. Generated column calculations such as `target_gap`, `target_achievement_rate`, and `ROI_marketing_calc` are automatically performed by the database.

3.1.4. Presentation Class

Power BI Desktop connects directly to the PostgreSQL schema datasheet via the built-in PostgreSQL connector. The dashboard provides interactive visuals with filtering capabilities by channel, time period, and campaign, enabling business users to self-service analytics without requiring technical expertise.

3.2. Technologies and tools used

Table 3.1 lists all the technologies used in the project:

Table 3.1 – List of technologies and tools

Technology	Type	Role	Reasons for choosing
PostgreSQL 15	Relational database	Stage storage and data warehouse	Open source, ACID compliant, supports Windows functions.
Python 3.x	Programming languages	Main ETL script	The panda ecosystem, psycopg2, is robust.
pandas	Python library	Data processing and transformation	High performance with DataFrame API
psycopg2	Python library	Connect to PostgreSQL	Driver Python-PostgreSQL

python-dotenv	Python library	Configuration management	Separate the configuration from the code.
openpyxl	Python library	Read Excel files	.xlsx support does not require Microsoft Office.
Docker	Container platform	PostgreSQL packaging	Ensuring a consistent environment
Docker Compose	Orchestration	Service management	Simplify startup/stopping.
Power BI Desktop	BI tools	Dashboard & visualization	Direct connection to PostgreSQL
Microsoft Excel	Office tools	Data source	An .xlsx file with 37 columns.

3.3. Deployment Environment

The entire system is designed to run on a personal computer (localhost) using Docker Desktop. The docker-compose.yml file defines a single service called PostgreSQL 15 Alpine (lightweight image, suitable for a development environment), maps port 5432 to the host, and uses a Docker volume (pgdata) to ensure data is not lost when the container restarts.

This containerization approach ensures the project can run consistently across all operating systems (macOS, Windows, Linux) without requiring a direct PostgreSQL installation. The run_etl.sh script automates the entire process from container startup to ETL completion with a single command.

CHAPTER 4: DATA WAREHOUSE DESIGN

4.1. Design Method

The data warehouse is designed using Ralph Kimball's Dimensional Modeling method, employing the Star Schema model. This is the most popular method for designing data warehouses for BI systems due to its superior advantages: simple structure, optimized queries, ease of understanding for business users, and good integration with BI tools such as Power BI.

In the Star Schema model, a central fact table (fact_marketing_sales) contains metrics and foreign keys that link to eight surrounding dimension tables. Each dimension table describes a specific analytical dimension such as time, channel, campaign, or funnel stage.

Compared to the Snowflake Schema (which is more standardized), Star Schema optimizes query speed by reducing the number of JOIN operations required. This is especially important when Power BI performs complex queries in real time for users interacting with the dashboard.

4.2. Dimension Tables

The system has 8 dimension tables, each describing an aspect (dimension) in marketing data analysis:

Table 4.1 – Description of Dimension Tables

Table name	Master key	Description and key attributes
dim_date	full_date (PK)	The calendar table includes year, month, quarter, month_name, and year_month. It allows for analysis at multiple time levels.
dim_stage	stage_name (PK)	The stages of the marketing funnel are defined by stage_order (numbers 1-5) and stage_group (groups: Awareness, Consideration, Intent, Conversion).

dim_channel	channel_name (PK)	Distribution channels, categorized by channel_type, include Social, Digital commerce, Broadcast, Trade promotion, and Traditional trade.
dim_customer_segment	segment_name (PK)	Customer segments: Modern channels, e-commerce, etc.
dim_campaign	campaign_name (PK)	Marketing campaign with campaign_type (Seasonal, Year end, Back to school, Other).
dim_department	department_name (PK)	Department: Operations, Marketing, Sales, etc.
dim_machine	machine_name (PK)	Machine codes: A, B, C.
dim_information	information_name (PK)	Source of information: Facebook orders, complaints, etc.

Dimension tables use natural keys (actual names) as primary keys instead of surrogate keys (self-incrementing IDs). This approach is suitable for data with low cardinality (a small number of unique values) and simplifies JOINS in Power BI. All INSERTs use ON CONFLICT DO UPDATE (upsert) to ensure idempotency when re-running the ETL.

4.3. Central Fact Table

The fact_marketing_sales table is the heart of Star Schema, containing 31 columns including 9 foreign keys and 22 metrics:

Table 4.2 – Structure of the fact_marketing_sales table

Column / Column Group	Data type	Describe
order_id	BIGINT PK	Order number, primary key of the fact table.
full_date	FK → dim_date	Trading day
stage_name	FK → dim_stage	Marketing funnel stage
channel_name	FK → dim_channel	Distribution channel

information_name	FK → dim_information	Source of information
machine_name	FK → dim_machine	Production machine
department_name	FK → dim_department	Department
segment_name	FK → dim_customer_seg ment	Customer segmentation
campaign_name	FK → dim_campaign	Marketing campaign
quantity, revenue, cost, profit, profit_margin	NUMERIC	Core sales figures
inventory_level, calls, waiting_time, conversion_proxy	NUMERIC	Performance indicators and funnel
budget, target, marketing_cost, roi_marketing	NUMERIC	Marketing budget and effectiveness index
raw_material_cost, labor_cost, logistics_cost, unit_cost, revenue_per_unit, cost_per_machine_unit	NUMERIC	Detailed cost analysis
market_size, market_share, region_profit_share	NUMERIC	Market Index
target_gap (GENERATED)	NUMERIC	Revenue - Target, calculated automatically by the database.
target_achievement_rate (GENERATED)	NUMERIC	Revenue / Target, calculated automatically by DB.
roi_marketing_calc (GENERATED)	NUMERIC	Profit / Budget, calculated automatically by DB.

A unique feature of the fact table is its three columns that are GENERALIZED ALWAYS AS STORED: target_gap (revenue gap from target), target_achievement_rate (target completion rate), and roi_marketing_calc (marketing ROI calculated from profit/budget). These columns are automatically calculated and

updated by PostgreSQL every time an INSERT or UPDATE is performed, ensuring consistency without requiring computational logic in ETL or Power BI.

4.4. Relationship Diagram

All nine foreign keys in the fact table have explicit FOREIGN KEY (REFERENCES) constraints to their respective dimension tables. The data loading order must be followed: dimension tables are loaded first (`load_dimensions()`), followed by the fact table (`load_fact()`). If a foreign key value does not exist in the dimension, the system will report a constraint violation; this is a crucial data quality check mechanism.

The fact table lacks a foreign key to the `dim_date` by region/branch because geographic dimensions are not modeled as separate dimensions in the current design. Region and branch information is held in the staging table but not included in the data warehouse because there is no corresponding dimension table – this is an area that could be improved in a future version.

CHAPTER 5: ETL PROCESS

5.1. Overview of ETL Pipeline

ETL (Extract-Transform-Load) is the core process of any data warehouse system. The `py-marketing.py` script implements the entire ETL pipeline in a single Python file, using `psycopg2` for database connection and `pandas` for data processing. The `.env` file contains all connection configuration and table names, completely separate from the code according to the 12-Factor App principle.

The pipeline is organized into four sequential steps in the `main()` function: (1) Connect to PostgreSQL, (2) Load Excel into staging (`load_to_staging()` function), (3) Read data from staging and clean it (`clean_data()` function), (4) Load into Data Warehouse in the following order: dimensions first (`load_dimensions()`), fact table second (`load_fact()`). The entire pipeline is enclosed in a single database connection and each step commits separately.

5.2. Extraction Stage

The Extract step reads the Excel file `masan_case.xlsx` using pandas with `dtype=str`, meaning all values are read as character strings. This is the safest strategy when loading into staging, preventing pandas from arbitrarily converting data types and losing information (for example, numbers with leading zeros, dates being incorrectly parsed).

After reading, all pandas' NaN values are converted to Python's None so that `psycopg2` can load them into PostgreSQL as NULL. The staging table is TRUNCATED before each load to ensure the staging always accurately reflects the current Excel file content, and then all the data is INSERT in bulk using `execute_values()` - the most efficient method for bulk inserts with `psycopg2`.

5.3. Transform Phase

5.3.1. Column Name Standardization

The ``standardize_columns()`` function addresses the issue of inconsistent column names across data versions. Each target attribute is mapped from multiple possible source name variations (e.g., "ROI_Marketing", "roi marketing", "roimarketing" are all recognized and changed to "roi_marketing"). This mechanism uses the ``normalize_col()`` function to remove any special characters and convert them to lowercase, then compares the normalized string, ensuring correct mapping regardless of how the user enters the column name.

5.3.2. Date Parsing

The `parse_dates()` function handles three different date formats: ISO `yyyy-mm-dd` (highest priority, no ambiguity), `dd/mm/yyyy` (common in Vietnam), and `mm/dd/yyyy` (fallback). The strategy of prioritizing ISO first ensures that dates like "2024-01-02" are parsed as January 2nd instead of February 1st (which a day-first parser might produce). The result is a Python date object, directly compatible with the PostgreSQL DATE type.

5.3.3. Cleaning text data

Text columns are striped (using ``strip()``) to remove leading/trailing whitespace, and null values (`""`), the string "nan", and "None" (which appear when pandas converts to string) are replaced with pandas NA. This ensures there are no spurious empty strings in the database, only genuine NULL values – crucial for the accuracy of aggregate calculations.

5.3.4. Converting number types and removing duplicates

All 22 numeric columns are converted using ``pd.to_numeric(errors="coerce")``, meaning that non-convertible values (strange characters, empty cells) will become NaN instead of causing errors. Rows missing values in the 5 required columns (order_id, date, channel, stage, revenue) will be dropped. Finally, ``drop_duplicates(subset=["order_id"], keep="last")`` ensures that each order appears only once in the fact table, retaining the most recent record.

5.4. Loading Phase

5.4.1. Loading the Dimension table (load_dimensions())

Eight dimension tables are loaded sequentially in a transaction, committed once after all are complete. Each dimension is extracted from the cleaned DataFrame by using ``drop_duplicates()`` on the primary key column, then INSERT with ``ON CONFLICT DO UPDATE (upsert)``. This ensures both scenarios: inserting a new dimension and updating an existing dimension if the property changes.

Dimension dim_stage is independent of the source data for order and grouping – these values are hardcoded in the STAGE_ORDER dictionary according to business logic (Impression=1/Awareness, Purchase=5/Conversion). Similarly, dim_channel uses CHANNEL_TYPE to categorize channels. This ensures consistency of dimension data even if the data source changes its spelling.

5.4.2. Loading the Fact table (load_fact())

The fact table is loaded after all dimensions, ensuring the FK constraint is always satisfied. INSERT also uses ON CONFLICT (order_id) DO UPDATE with 30

columns updated. This upsert approach allows the ETL to be run multiple times without creating duplicate data – an important idempotency for a production pipeline.

5.5. Error handling and quality assurance

The script uses the context manager "with psycopg2.connect() as conn:" to automatically rollback the transaction if an exception occurs, preventing the database from being in an inconsistent state. The `execute_values_if_any()` function checks that the row list is not empty before calling `execute_values()`, avoiding SQL syntax errors when the dimension has no data. Finally, the `clean_value()` function converts any NaN of pandas to None before passing it to SQL, avoiding the common "can't adapt type 'float'" error with psycopg2.

CHAPTER 6: DASHBOARD AND VISUALIZATION

6.1. Overall Design

The Power BI dashboard (file `pbi-marketing.pbix`) is built on the "single-page dashboard" philosophy – all important information is on a single page (Page 1), eliminating the need for scrolling. This design is optimized for quick monitoring and decision-making during executive meetings. The dashboard connects directly to the

PostgreSQL schema dw, ensuring that the data is always the latest version after each ETL run.

Power BI automatically builds a data model based on tables in the schema dw, identifying relationships through matching column names (full_date, stage_name, channel_name, etc.). Measures in Power BI (Revenue, Profit, Marketing ROI, Goal Acquisition Rate) are calculated using the DAX language and stored in a virtual _measures table.

6.2. Dashboard Components

Table 6.1 details the 10 visuals in the dashboard:

Table 6.1 – Description of visuals in pbi-marketing.pbix

Visual	Type of analysis	Data displayed	Analytical value
KPI Card (×4)	Overview	Revenue, Profit, Marketing ROI, Targets to be achieved	Provides a quick overview of 4 key metrics as soon as you open the dashboard.
Combo Chart (Column + Line)	Channel analysis	Revenue and Profit (columns) + Marketing ROI (lines) by channel_name	Visual comparison between the scale (columns) and efficiency (lines) of each channel.
Gauge	Track goals	Actual revenue compared to target	Visualize your progress toward achieving your goals using a progress meter.
Clustered Bar Chart	Channel efficiency	Target achievement rate (target_achievement_rate) by channel_name	Compare the relative effectiveness of channels based on the criterion of achieving objectives.

Funnel Chart	Funnel analysis	Revenue by stage_name (from Impression to Purchase)	Visualize dropout rates at each stage of the transition.
Treemap	Campaign	Revenue by campaign_name, cell size corresponds to the value.	Quickly identify the campaigns that generate the most revenue.
Slicer – Channel	Filter data	channel_name	Allows filtering the entire dashboard by one or more channels.
Slicer – Date Hierarchy	Time filtering	Year / Quarter / Month / Day	Flexible time period analysis
Slicer – Campaign	Filter campaigns	campaign_name	Focus the analysis on the specific campaign.

6.3. KPI System

The dashboard tracks four key KPIs through visual cards: (1) Revenue - total revenue; (2) Profit - total gross profit; (3) Marketing ROI - marketing return on investment, calculated as profit/budget; (4) Target - total target. These KPIs provide an immediate overview of business health when the dashboard is opened.

Index	Recipe	Meaning
Marketing Cost	Actual marketing costs	Allocated marketing expenses
Budget	Planned budget	Resources are expected to be allocated.
ROI on budget	Profit / Budget	Profit generated within the allocated budget.
Marketing cost efficiency	Profit / Marketing Cost	How much profit does one unit of marketing expenditure generate?
Marketing expense ratio	Marketing Cost / Revenue	What percentage of

		revenue is used for marketing?
Target achievement rate	Revenue / Target	Level of revenue target achievement

In addition to the four key KPIs, Gauge visual tracks goal progress in real time, while charts provide deeper analysis of performance by channel and campaign. Three Slicers allow business users to self-serve by filtering data by channel, time, and campaign without needing to contact the technical department.

6.4. Power BI Connection Guide

After the ETL is complete, follow these steps to connect Power BI to your data warehouse:

1. Open the pbi-marketing.pbix file in Power BI Desktop.
2. Vào menu Home → Transform data → Data source settings.
3. Select the PostgreSQL data source and click Change Source.
4. Enter the following information: Server = localhost, Database = bi_db.
5. Click OK, then Edit Credentials: User = bi_user, Password = bi_pass, Authentication = Database.
6. Click Connect, then Close & Apply to refresh the data.

CHAPTER 7: ANALYSIS OF RESULTS

7.1. Performance analysis by revenue

The data warehouse model allows for the effective analysis of six distribution channels from multiple perspectives. The combo chart in the dashboard simultaneously displays absolute revenue (columns) and Marketing ROI (lines), creating a balanced view of the contribution scale and investment efficiency of each channel.

The goal of the analysis is not only to identify the channel or campaign that generates the most revenue, but also to answer three key management questions. First, are the marketing resources used generating a commensurate return? Second, which channels and campaigns should be prioritized, maintained, or adjusted? Third, do the results achieved meet the business objectives?

7.2. The Overall Picture

Index	Result
Revenue	7.774.179,29
Profit	1.849.205,20
Marketing Cost	592.497,93
Budget	159.650.378,64
ROI on budget	2,32%
Profit/Marketing Cost	3.12 times
Marketing Cost/Revenue	7,75%
Target achievement rate	9,64%
Budget utilization rate	0,80%

Considering all the data, the business generated approximately 7.77 million units of revenue and 1.84 million units of profit. With 592,000 units of marketing expenses, each unit of marketing expense generated approximately 3.12 units of profit.

However, the ROI on the total budget was only 2.32%, and revenue reached only about 9.67% of the target. At the same time, actual marketing expenses were only equivalent to 0.80% of the recorded budget. The significant gap between Budget, Marketing Cost, and Target indicates that the business needs to re-examine its budgeting methods, measurement units, and target levels before making resource allocation decisions.

Therefore, the main problem is not that the business is not generating profits, but rather the imbalance between the budget, actual spending, and expected targets.

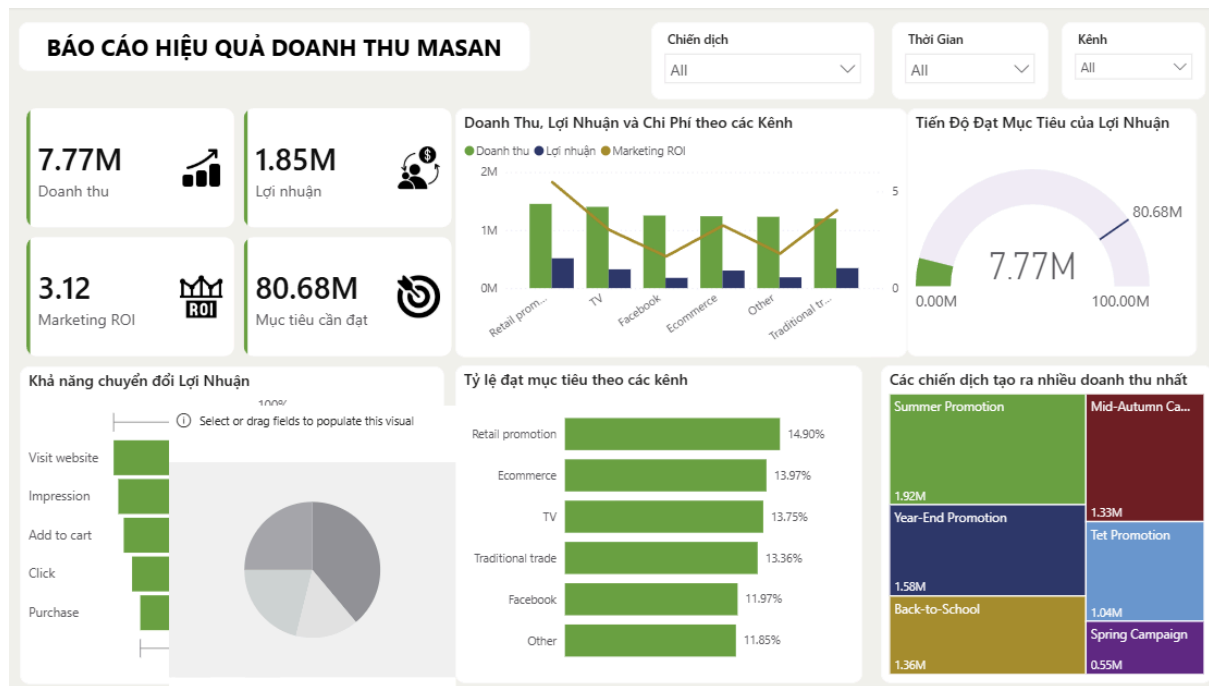


Figure 1. Revenue Performance Report

7.3. Conversion Funnel Analysis

The funnel chart in the dashboard visualizes the 5 conversion stages: Impression → Visit website → Click → Add to cart → Purchase. The star schema model allows for conversion rate analysis between stages through dim_stage with the stage_order and stage_group attributes.

The Impression (Awareness) stage typically generates the largest volume and revenue because it's the first point of contact. The conversion_proxy data in the fact table provides a quantitative estimate of the conversion rate at each step. Comparing funnels by channel (Facebook vs TV vs Ecommerce) allows for optimized marketing budget allocation – investing more in the channel with the best conversion rate from the interest stage to purchase.

Khả năng chuyển đổi Lợi Nhuận



Figure 2. Profitability

7.4. Revenue Channel Analysis

Channel	Revenue	Profit	Marketing ROI	Target achievement rate
Retail promotion	1.452.388,88	514.093,01	5.48 times	14,90%
Traditional trade	1.199.960,26	344.546,72	4.03 times	13,36%
Ecommerce	1.239.288,86	303.300,76	3.24 times	13,97%

TV	1.401.337,35	324.141,80	3.01 times	13,75%
Other	1.229.611,26	186.545,10	1.79 times	11,85%
Facebook	1.251.592,68	176.577,81	1.64 times	11,97%

Retail promotion is the most prominent channel, simultaneously leading in revenue, profit, ROI on budget, and marketing cost performance. One unit of marketing expenditure in this channel generates approximately 5.48 units of profit.

Traditional trade and e-commerce did not generate the highest revenue, but they performed relatively well, producing 4.03 and 3.24 units of profit per unit of marketing expenditure, respectively.

Conversely, Facebook is the least effective channel. Marketing costs account for 9.22% of revenue, but each unit of marketing cost only generates 0.85 units of profit. This indicates that the profit generated is not enough to directly offset the channel's marketing costs.

TV generates relatively large revenue, but its profit margin is lower than retail promotion, traditional trade, and e-commerce. Therefore, high revenue does not necessarily mean high return on investment.

Tỷ lệ đạt mục tiêu theo các kênh

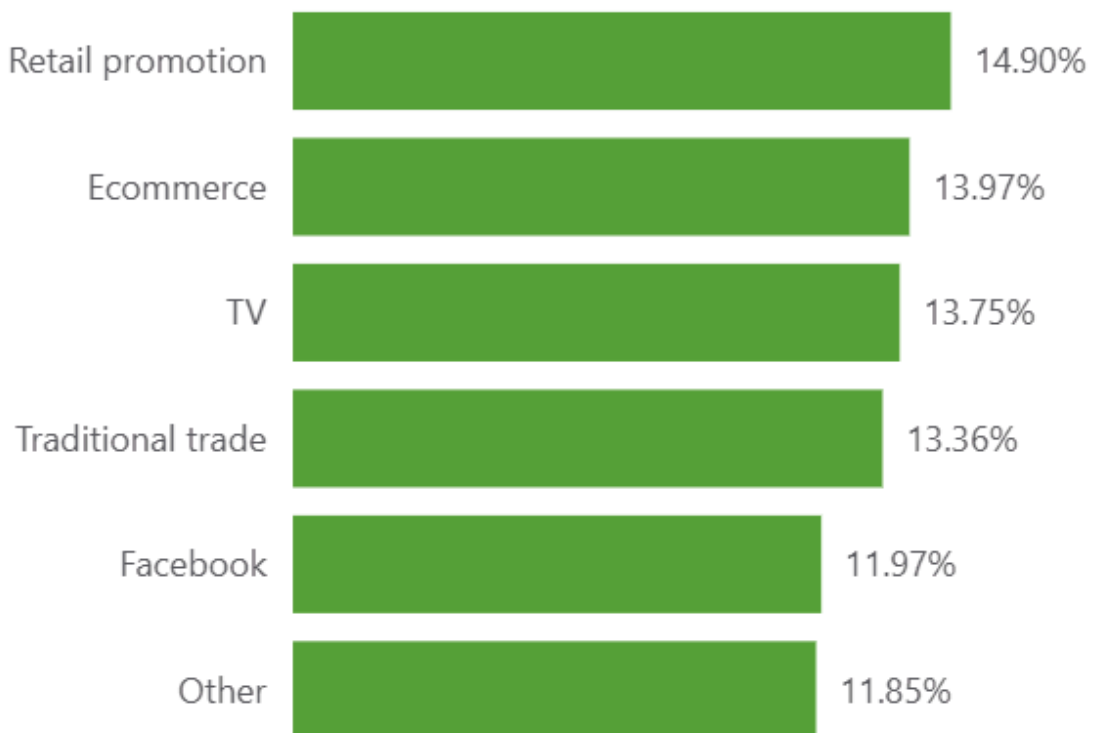


Figure 3. Statistics of items by channel

7.5. Revenue analysis by campaign

Campaign	Revenue	Revenue share
Summer Promotion	1.916.731,65	24,66%
Year-End Promotion	1.580.546,99	20,33%
Back-to-School	1.355.463,12	17,44%

Mid-Autumn Campaign	1.330.269,46	17,11%
Tet Promotion	1.037.480,79	13,35%
Spring Campaign	553.687,28	7,12%

The Summer Promotion generated the highest revenue, but it was also the least effective campaign. Each unit of marketing expenditure only generated 0.54 units of profit, while the ROI on the budget was only 0.47%.

Conversely, the Mid-Autumn Campaign didn't lead in revenue but performed best, generating 5.86 units of profit per unit of marketing expenditure. The Year-End Promotion also achieved a good balance between revenue size and effectiveness.

This result illustrates an important management principle: campaigns should not be judged solely on revenue. A campaign that generates high revenue but incurs high costs and produces little profit may be less valuable than a campaign with lower revenue but a higher ROI.

Các chiến dịch tạo ra nhiều doanh thu   ...

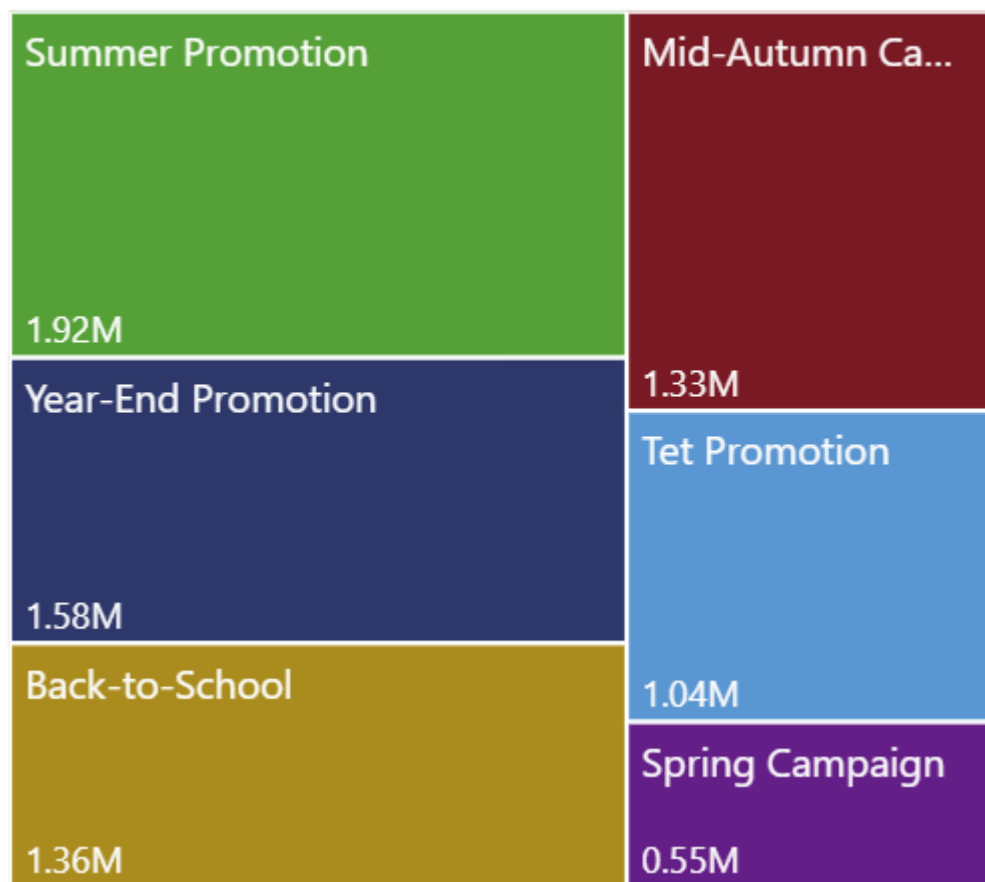


Figure 4. Revenue-generating campaigns

7.6. Conclusion on budget management and allocation

Decision group	Object
Expansion should be prioritized.	Retail promotion, Mid-Autumn, Year-End
It should be maintained and optimized.	Ecommerce, Traditional trade, Back-to-School
Redesign is needed.	Facebook, Summer Promotion, Tet Promotion

Further monitoring of the identification role is needed.	TV, Impression
Data quality needs to be checked.	Budget, Target, ConversionProxy

Based on the analysis, marketing decisions should not be based solely on revenue. Businesses need to simultaneously evaluate revenue size, profit, marketing costs, ROI, and the extent to which objectives are being achieved.

Retail promotion, Mid-Autumn Campaign, and Year-End Promotion are suitable targets for prioritizing resources due to their scale and effectiveness. Conversely, Facebook and Summer Promotion need to be redesigned before further budget expansion.

However, the unusually low budget utilization rate and target achievement rate suggest that the measurement units and logic behind the Budget and Target framework need to be re-examined before applying the recommendations in practice.

7.7. Multi-dimensional Analysis

The true power of this BI system lies in its interactive, multi-dimensional analytical capabilities. For example, an analyst can select the slicer "Channel = Facebook" + "Year = 2024" + "Campaign = Year-End" to see the exact performance of the 2024 Year-End campaign on Facebook, including revenue, ROI, and target progress. All 10 visuals update simultaneously and consistently. This is the core value that the BI system brings compared to traditional static Excel reports.

CHAPTER 8: CONCLUSION AND RECOMMENDATIONS

8.1. Conclusion

The project successfully built a complete end-to-end Business Intelligence system for Masan Group's marketing data analysis. The system includes: an automated ETL process (py-marketing.py) that converts raw Excel data into a PostgreSQL data warehouse; a Star Schema-based data warehouse with one fact table and eight dimension tables; and an interactive Power BI dashboard with 10 multi-dimensional marketing analytics visualizations.

The initial goals were all achieved: (1) ETL automation with a single, iterable Python script (idempotent thanks to upsert); (2) A well-designed data warehouse based on dimensional modeling standards; (3) An intuitive dashboard providing a 360-degree view of marketing effectiveness; (4) Containerization with Docker ensuring portability and environment consistency; (5) All configurations managed via .env, not hardcoded in code.

Technically, the system demonstrates many best practices: robust column name normalization across multiple input variations; date parsing that handles all three common formats; upsert instead of simple insert; generated columns for consistency; and a clear, modular code structure.

8.2. Limitations of the project

Despite achieving its objectives, the project still has some limitations that need to be acknowledged. Firstly, the current system is full-refresh (TRUNCATE and reload the entire staging area each time), suitable for small datasets but inefficient when dealing with millions of records – requiring an incremental loading approach. Secondly, there is no logging or alerting mechanism for ETL errors in the production environment. Thirdly, geographic dimensions (Region, Branch) have not been modeled as separate dimensions, limiting the ability to perform geographic-based analysis. Fourthly, the dashboard currently only has one page and has not been divided into specialized pages such as Overview, Channel Analysis, and Campaign Analysis.

8.3. Development direction

The project has significant potential for future expansion. From an ETL perspective, it's possible to implement incremental loading based on watermarks (date or order_id), integrate Apache Airflow for automated pipeline scheduling and monitoring, add data quality checks using the Great Expectations library, and integrate with other data sources (CRM, ERP, Google Analytics).

In terms of data modeling, you can add dim_geography (province/city, region) for geographic analysis, more detailed dim_product with category hierarchy, and new fact tables like fact_inventory or fact_customer_service to expand the scope of analysis. In terms of analytics, you can integrate Python scripts into Power BI to run revenue and budget forecasting models, or implement a REST API so other systems can query the data warehouse in real time.

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